

Gustavo Garcia

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EDUCATION

University of California San Diego

Bachelor of Science in Mathematics-Computer Science

La Jolla, CA

Sep 2024 - June 2028

EXPERIENCE

Triton Droids - Vice President of Engineering

La Jolla, CA | Jan 2025 – Present

- Trained reference-motion and velocity commanded PPO locomotion policies in MJLab/MuJoCo for a custom humanoid and deployed a 50 Hz ONNX policy using IMU/joint feedback and CAN motor commands.
- Built a MuJoCo system identification pipeline to fit actuator inertia, friction, damping, and delay from hardware step/sine tests. Moving from sim -> real, to sim -> real -> better sim -> real.
- Diagnosed sim-to-real gait instability across actuator response, command slew, and feedback timing instrumented the hardware control loop and aligned simulator inputs with processed motor targets.
- Lead engineering across a 100 member humanoid robotics club and two humanoid generations, secured sponsorships from large robotics companies to empower the next generation of robotics engineers.

PROJECTS

OpenRECAP - Self Improving Vision Language Action Model

Python, Pytorch, MuJoCo, LeRobot, Reinforcement Learning

- Built an open source, single GPU alternative to a closed, ~3B-parameter VLA offline RL approach by replacing large VLM based value estimation with a compact state conditioned MLP.
- Built an offline RL pipeline for SmolVLA with SO-101 simulation, demonstrations, autonomous rollouts, reward labeling, and advantage weighted fine tuning, deployed on physical SO-101 arms.
- Improved task success from 50% to 60% with dense rewards, raised success on the hardest object placement from 20% to 51%.

Humanoid Robot Fighting - Motion Imitation Learning

Python, MuJoCo, MJLab, PPO, RSL-RL

- Built motion retargeting pipeline for EngineAI's T800 robot, mapping 30 Hz BVH/NPY boxing clips to 50 Hz trajectories across 25 actuated joints using FK and quaternion SLERP.
- Adapted PPO motion tracking for 14 body segments with pose, velocity, joint limit, action rate, and self collision objectives.
- Trained punch/kick imitation policies and qualified in the top 32 of the Ultimate Robot Knockout League's opening round.

Vision Language Action Models for Manipulation

Python, LeRobot, Action Chunking Transformers

- Benchmarked and fine tuned $\pi 0.5$, ACT, and SmolVLA on cube stacking across varied object positions, $\pi 0.5$ achieved 10% task success versus 0% for both baselines.
- Fine-tuned $\pi 0.5$ for 50K steps and deployed VLA policies on physical SO-101 arms, traced real world failures to camera pose, lighting, and teleoperation distribution shift.
- Ran H100 training experiments across VESSL AI, Colab, and SDSC tracked runs in Weights & Biases and published model checkpoints to Hugging Face

SKILLS

Languages & Tools: Python, C++, MATLAB, PyTorch, LeRobot, MuJoCo, Isaac Lab, HuggingFace, Git, Linux/Ubuntu, Onshape, EagleCAD

Models & Methods: Reinforcement learning, $\pi 0.5$, ACT, SmolVLA, PPO, SAC, Flow Matching, Behavior Cloning, Teleoperation, CAD/Mechanical Design